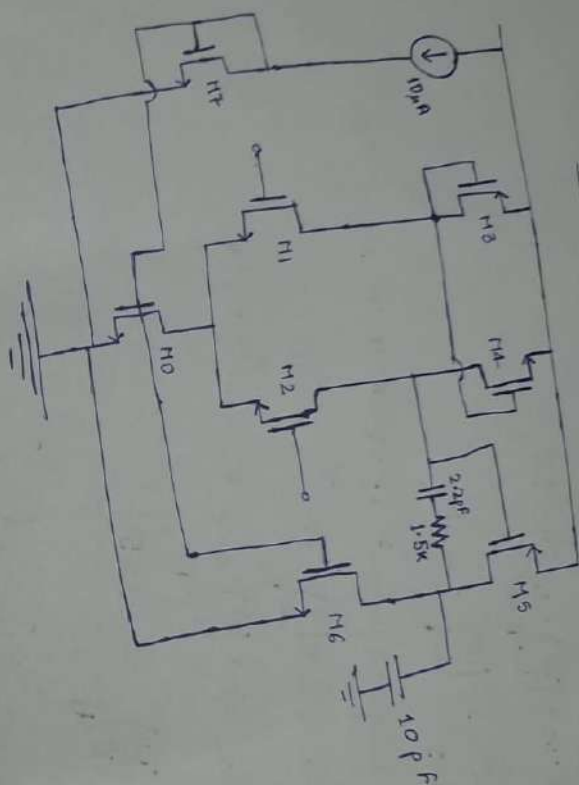


Hand Calculations



Given:

$$V_{tn} = 0.57V$$

$$V_{tp} = 0.39V$$

$$\mu_n C_{ox} = 230 \mu A/V^2$$

$$\mu_p C_{ox} = 100 \mu A/V^2$$

$$V_{dd} = 1.8V$$

$$L_{min} = 0.18 \mu m$$

$$W_{min} = 0.27 \mu m$$

$$V_{ov} = 0.2V$$

Assuming a tail current of $100 \mu A$,

$$\frac{1}{2} \times \mu_n C_{ox} \times \left(\frac{W}{L}\right)_0 \times (V_{ov})^2 = 100$$

$$\Rightarrow \frac{1}{2} \times 230 \times \left(\frac{W}{L}\right)_0 \times 0.04 = 100$$

$$\Rightarrow \left(\frac{W}{L}\right)_0 = 21.74$$

$$\frac{1}{2} \times \mu_n C_{ox} \times \left(\frac{W}{L}\right)_n \times (V_{ov})^2 = 50$$

$$\Rightarrow \left(\frac{W}{L}\right)_1 = \left(\frac{W}{L}\right)_2 = 10.87$$

$$\frac{1}{2} \mu_p C_{ox} \times \left(\frac{W}{L}\right)_p \times (V_{ov})^2 = 1050$$

$$\Rightarrow \frac{1}{2} \times 100 \times \left(\frac{W}{L}\right)_p \times 0.04 = 50$$

$$\Rightarrow \left(\frac{W}{L}\right)_3 = \left(\frac{W}{L}\right)_4 = 25$$

Assuming a current of $250 \mu A$ for the second stage.

$$\frac{1}{2} \times \mu_p C_{ox} \times \left(\frac{W}{L}\right)_p \times (V_{ov})^2 = 250$$

$$\Rightarrow \frac{1}{2} \times 100 \times \left(\frac{W}{L}\right)_p \times 0.04 = 250 \times 250$$

$$\Rightarrow \left(\frac{W}{L}\right)_5 = 125$$

$$\frac{1}{2} \times \mu_n C_{ox} \times \left(\frac{W}{L}\right)_n \times (V_{ov})^2 = 250$$

$$\Rightarrow \left(\frac{W}{L}\right)_6 = 54.345$$

	M_0	M_1	M_2	M_3	M_4	M_5	M_6	M_7
g_m	$\frac{1}{1k\Omega}$	$\frac{1}{2k\Omega}$	$\frac{1}{2k\Omega}$	$\frac{1}{2k\Omega}$	$\frac{1}{2k\Omega}$	$\frac{1}{400\Omega}$	$\frac{1}{400\Omega}$	$\frac{1}{10k\Omega}$
r_o	$50k\Omega$	$100k\Omega$	$100k\Omega$	$133.33k\Omega$	$133.3k\Omega$	$26.67k\Omega$	$20k\Omega$	$500k\Omega$
$V_{GS} - V_T$	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
I_D	$100\mu A$	$50\mu A$	$50\mu A$	$50\mu A$	$50\mu A$	$250\mu A$	$250\mu A$	$10\mu A$

Assuming $\lambda_n = 0.2 V^{-1}$ $\lambda_p = 0.15 V^{-1}$

Theoretical gain:
$$\frac{1}{2k\Omega} \times 57.14k\Omega \times \frac{1}{0.4k\Omega} \times 11.43k\Omega$$

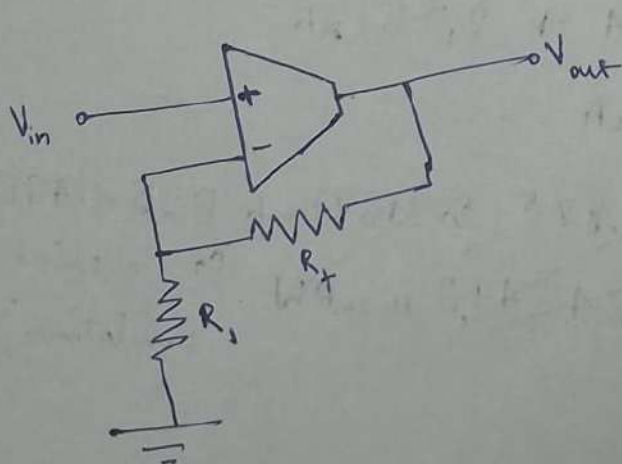
$$= 816.38 \approx 58.23 \text{ dB}$$

Simulated Results

	M_0	M_1	M_2	M_3	M_4	M_5	M_6	M_7
I_D (μA)	100 μA	50.47 μA	50.47 μA	50.47 μA	50.47 μA	236 μA	236 μA	10 μA
$V_{gs} - V_{Th}$ (V)	0.2621	0.263	0.263	0.2271	0.2271	0.23	0.221	0.221
g_m (Ω^{-1})	$\frac{1}{1.105 k\Omega}$	$\frac{1}{2.605 k\Omega}$	$\frac{1}{2.605 k\Omega}$	$\frac{1}{2.25 k\Omega}$	$\frac{1}{2.25 k\Omega}$	$\frac{1}{487 \Omega}$	$\frac{1}{468 \Omega}$	$\frac{1}{11.05 k\Omega}$
r_o ($k\Omega$)	50 $k\Omega$	99.06 $k\Omega$	99.06 $k\Omega$	132.09 $k\Omega$	132.09 $k\Omega$	28.25 $k\Omega$	21.18 $k\Omega$	500 $k\Omega$

Practical gain : 56.627 dB

Closed loop Gain :



$$R_f = R_1$$

$$\text{Gain} = 1 + \frac{R_f}{R_1} = 2 \approx 6.02 \text{ dB}$$

Values of other components :

- i) load capacitance $(C_L) = 10 \text{ pF}$
- ii) Miller capacitance $(C_c) = 2.2 \text{ pF}$
- iii) Feedback resistance $(R_F) = 10 \text{ k}\Omega$
- iv) $R_1 = 10 \text{ k}\Omega$
- v) Value of ideal current source $= 10 \mu\text{A}$

Power Consumption :

Theoretical — $V_{DD} \times I_{\text{total}}$

$$I_{\text{total}} = (10 + 100 + 250) \mu\text{A} = 360 \mu\text{A}$$

$$\therefore \text{Power consumed} = 1.8 \text{ V} \times 0.36 \text{ mA} \\ = 648 \mu\text{W}$$

Practical — $V_{DD} \times I_{\text{total}}$

$$\cancel{I_{\text{total}} = (10 + 100.94 + 2.56) \mu\text{A}} \\ \cancel{= 346.94 \mu\text{A}}$$

$$\therefore \text{Power consumed} = \cancel{1.8 \text{ V} \times 0.34694 \text{ mA}} \quad 625.04685 \mu\text{W} \\ \cancel{= 624.492 \mu\text{W}} \quad (\text{as obtained from simulation})$$